

Data communications (Lab 10): Nyquist Sampling Theorem

Objective: To demonstrate the Nyquist Sampling Theorem using MATLAB by plotting:

- Original continuous sine wave
- Signal sampled at Nyquist frequency
- Signal sampled below Nyquist frequency and observe distortion (aliasing)

Theory

According to Nyquist Sampling Theorem, a continuous-time signal can be accurately reconstructed if it is sampled at a frequency at least twice the highest frequency component of the original signal.

Mathematically:

$$F_s \geq 2F_m$$

where:

F_s = sampling frequency

F_m = maximum frequency of the original signal.

The quantity: $2F_m$ is called the Nyquist Rate.

Case 1: Proper Sampling

If $F_s \geq 2F_m$ then:

- enough samples are taken
- original signal can be reconstructed accurately
- no distortion occurs

Case 1: Under sampling

If $F_s < 2F_m$ then:

- insufficient samples are taken.
- signal becomes distorted,
- aliasing.

Example: Original Signal: $x(t) = A \sin(2\pi ft)$

$$f = 10 \text{ Hz}$$

$$\text{Nyquist frequency: } F_s = 2f = 20 \text{ Hz}$$

Sampling below Nyquist: $F_s = \frac{f}{4} = 2.5 \text{ Hz}$, this causes distortion.

MATLAB Code

Your task is to write a Matlab code that demonstrate the Nyquist rate Properties:

- Under sampling
- Oversampling

Sample Output:

